MD. HASAN IMON

AI & ML Engineer

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SUMMARY

Aspiring AI/ML Engineer with hands-on expertise in Generative AI, Agentic AI, and Machine Learning. Skilled in building production-ready solutions including multi-agent systems, RAG pipelines, and memory-augmented AI. Experienced in fine-tuning large language models (LLMs) using PEFT, LoRA, and QLoRA. Strong background in NLP (text classification, sentiment analysis, transformers, and language modeling) with a track record of delivering intelligent, tool-augmented pipelines that feature advanced reasoning, planning, and decision-making. Passionate about developing scalable, real-world AI systems that create measurable business impact.

PROJECTS

TrueWealth AI: Your AI-powered financial strategist – GitHub

Multi-agent financial advisor with Query breakdown & multi-stage planning.

- Developed a multi-agent AI pipeline to analyze and synthesize financial information from multiple sources for structured financial reasoning. Leveraged dynamic Tool Router and multi-source reasoning for accurate guide.
- Enhanced with persistent memory & multi-turn reasoning for continuous financial context tracking. Designed a modular, real-time web interface for user-friendly financial decision-making. Dockerized & deployed on Render with CI/CD pipelines ensuring seamless updates.

InformaTruth: AI-Driven News Authenticity Analyzer – GitHub

Build explainable fake news detection system using Fine-tuned models.

- Fine-tuned RoBERTa (classification) and Flan-T5 (explanation) for predictive with interpretable outputs. Integrated dynamic Tool Router + fallback agents (DuckDuckGo, Wikipedia) for source verification.
- Context-aware predictions, ensuring higher accuracy and reliability. Dynamic Tool Router and fallback agents to retrieve verified sources from DuckDuckGo and Wikipedia. Designed transparent, user-friendly outputs that public trust, ensuring explainable AI for news evaluation.

AutoDocThinker: Intelligent Agentic Search Engine – GitHub

Build an intelligent document QA system using Agentic RAG

- Developed a multi-format ingestion pipeline with database storage, enabling the system to parse, chunk, and semantically retrieve knowledge from heterogeneous documents with high accuracy and scalability.
- Implemented dynamic Tool Routing and fallback strategies that adaptively select the most reliable information source (RAG, Web Search, or Memory) based on context, ensuring trustworthy and resilient information delivery. Designed a user interface with CI/CD enabling seamless interaction with documents of any format.

Translatica: English to Spanish Translation – GitHub

Developed a Fine tuned (PEFT + LoRA) for high-fidelity translator with CPU-optimized deployment.

- Preserving nuanced tone, stylistic elements, and contextual meaning. Integrated evaluation metrics (BLEU, METEOR, and human evaluation) to ensure translation quality and consistency across diverse literary genres.
- Evaluated with BLEU, METEOR, and human assessments for translation quality. Dockerized & CI/CD enabled, ensuring smooth updates and lightweight CPU inference. Built a Flask-based modular web app, deployed on Render for real-time user translation.

MediGenius: Medical AI Assistant – GitHub

Build an AI-driven medical assistant with Multi-agent communication control.

- Multi-agent medical AI assistant orchestrated with LangGraph, enabling dynamic communication between agent to agent protocol. Integrated short-term memory with state-based reasoning for doctor-like, empathetic responses from medical PDFs, research papers, and fallback web sources like various website.
- Context-aware, multi-turn consultations. Designed an accessible, responsive web interface optimized for low latency, accurate guidance. Dockerized & deployed with CI/CD pipelines on Render for ensuring reliability.

SKILLS

- ✓ **Programming & Data:** Python, R, Data Analytics, Pandas, NumPy, Matplotlib, Seaborn.
- ✓ **MLOps & Backend:** REST APIs, FastAPI, Flask, Git/GitHub, OOP, Docker, Kubernetes, Postman
- ✓ **Machine Learning & Deep Learning:** Supervised & Unsupervised Learning, Classification, Regression, Clustering, Model Evaluation, PyTorch, TensorFlow, Keras, Scikit-learn
- ✓ **NLP & Large Language Models (LLMs):** Text Classification, Embeddings, Hugging Face, NLTK, spaCy, HuggingFace Transformers, GPT-3.5/4, LLaMA, Mistral, Claude, Custom Fine-tuning (PEFT, LoRA, QLoRA)
- ✓ **AI Agents & Multi-Agent Systems:** LangChain, LangGraph, AgentOps, CrewAI, LlamaIndex, Agentic RAG Pipelines, Agent Orchestration, Tool-Augmented, Memory-Augmented, Prompt Engineering, HITL
- ✓ **Vector Databases & Storage:** FAISS, ChromaDB, Pinecone, Weaviate, MySQL, MongoDB, PostgreSQL
- ✓ **Cloud & Deployment:** Docker, Kubernetes, AWS (EC2, S3, Lambda basics), CI/CD pipelines

EDUCATION

Bachelor of Computer Science & Engineering

City University, Bangladesh (2022 - Present)

CERTIFICATION

- **Machine Learning Specialization** by Stanford University (Coursera)
- **Programming for Everybody** by University of Michigan (Coursera)